

Yeshvanth Kumar Domala

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Education

California State University, East Bay
Master of Science in Computer Science

Aug 2024 – May 2026
Hayward, CA

Technical Skills

Languages: Python, Java, C++, TypeScript, JavaScript, SQL, Bash
Distributed Systems & Backend: Apache Spark, Kafka, ETL Pipelines, FastAPI, Node.js, REST APIs
Cloud & DevOps: AWS (Glue, Athena, Lambda, S3, Bedrock, DynamoDB, CDK), Docker, Kubernetes, Terraform, CI/CD
AI/ML: PyTorch, LangChain, RAG, FAISS, OpenCV, YOLOv8, MCP
Databases: PostgreSQL, MySQL, MongoDB, Redis
Frontend & Tools: React, HTML/CSS, GraphQL, Git, Linux

Work Experience

California State University, East Bay
Graduate Research Assistant – Applied ML & Robotics

Jan 2025 – Present
Hayward, CA

- Accelerated robotic perception by 38% through CUDA optimization on NVIDIA Jetson platforms, enabling real-time inference for autonomous robotic navigation.
- Developed telemetry visualization and evaluation pipelines for LLM-enabled robotic systems using ROS2 and PyTorch, improving experiment reproducibility and streamlining debugging workflows.

Amazon Web Services (AWS) – Aurora
Software Development Engineer Intern

May 2025 – Aug 2025
Redmond, WA

- Designed and deployed a distributed telemetry ETL platform processing 4–6 TB/hour of Amazon Aurora production logs using AWS Glue, Lambda, Athena, and S3, reducing manual investigation time by 90%.
- Engineered a schema-agnostic PySpark framework parsing 40–50 dynamic telemetry metrics into optimized Parquet and Amazon Ion datasets, reducing Athena storage costs by 75% through partition projection and Zstandard compression.
- Integrated Model Context Protocol (MCP) tooling to enable natural-language querying of production telemetry datasets, simplifying Athena SQL generation and improving engineering productivity.

Saama
Software Engineer

Jul 2023 – Jul 2024
Pune, India

- Engineered cloud-native data pipelines using Snowflake, StreamSets, and SQL for enterprise insurance datasets, supporting scalable analytics and financial reporting workflows.
- Built reusable data transformation and validation components, reducing pipeline execution time by 35% while improving downstream reporting reliability across multiple business teams.

Defense Research & Development Organization (DRDO)
Software Engineer Intern

May 2023 – Jul 2023
Hyderabad, India

- Improved backend throughput by 20% under high-concurrency workloads using asynchronous I/O and connection pooling in Node.js and C++.
- Implemented centralized ELK-stack logging and automated testing pipelines, improving failure detection and reducing production incidents.

Projects

Atlas AI – RAG Knowledge Platform | Python, FastAPI, React, FAISS, Docker, AWS

- Built a full-stack RAG platform enabling natural-language search across document collections with citation-aware responses; deployed on AWS with Docker, improving retrieval accuracy by 22% over a dense-only baseline through hybrid dense-sparse search.

SkyPilot AI – Autonomous Drone Mission Platform | Python, ROS2, FastAPI, AWS IoT Core, Amazon Bedrock, YOLOv8

- Developed an autonomous drone mission platform integrating Amazon Bedrock, ROS2, AWS IoT Core, and YOLOv8 for natural-language mission planning, real-time object detection, and cloud telemetry.
- Achieved 93% object detection accuracy with sub-250 ms end-to-end telemetry latency in simulation.

Leadership

Girl Up – United Nations Foundation
Head of Research & Documentation

2020 – 2023
Remote

- Led research and technical documentation for 15+ STEM initiatives across universities, developing AI/ML educational content that reached 500+ students and tripled workshop participation over two years.